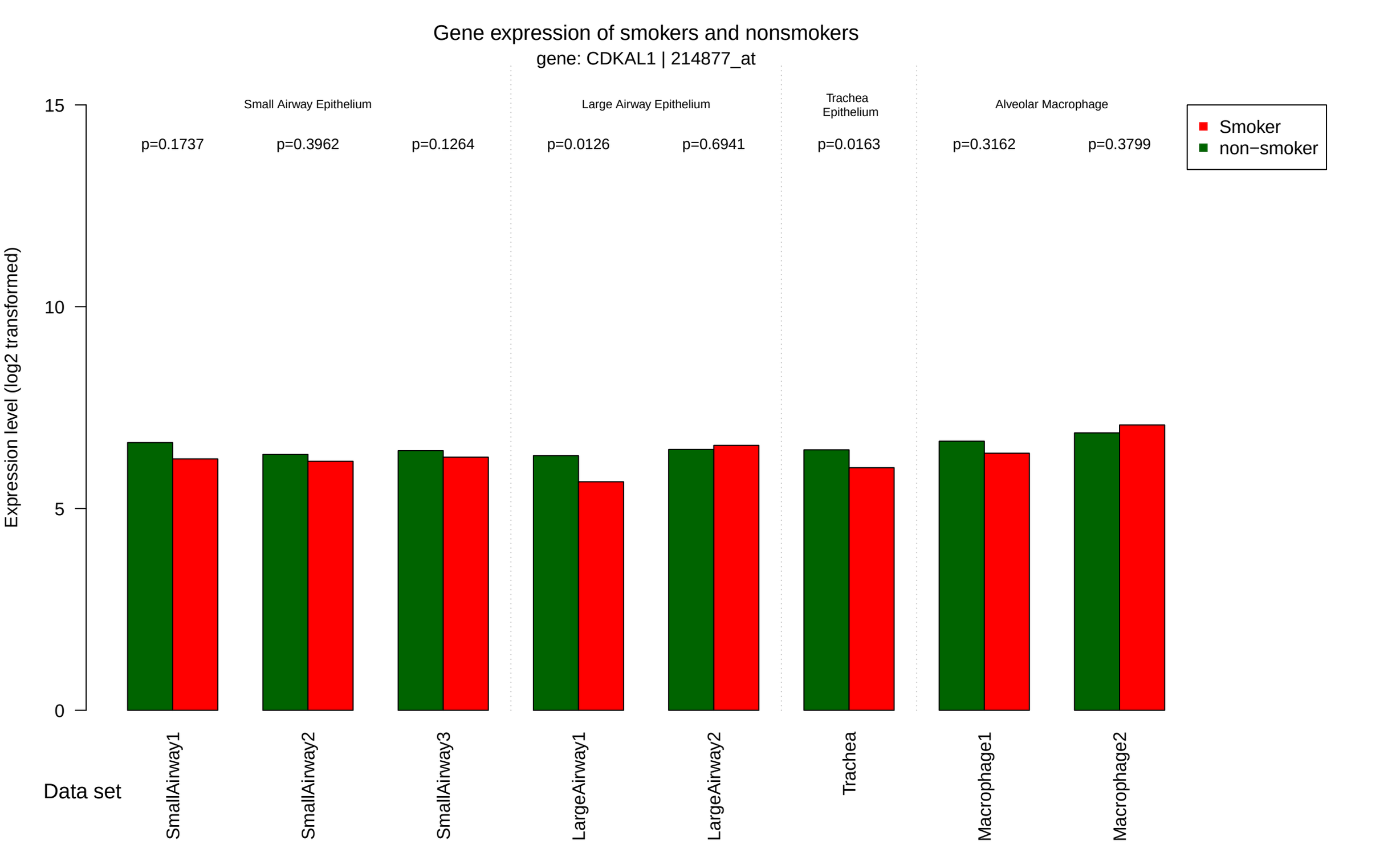
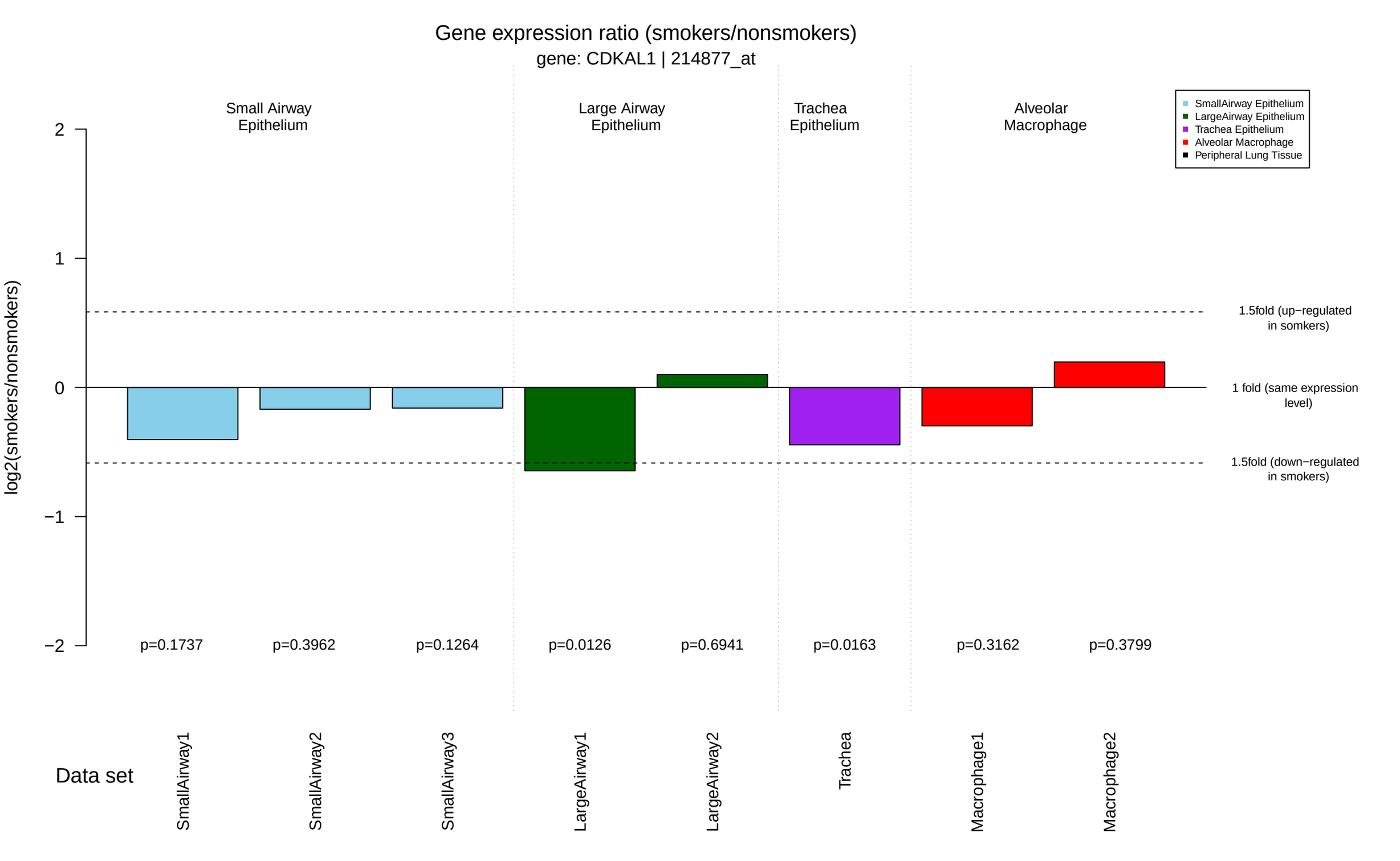
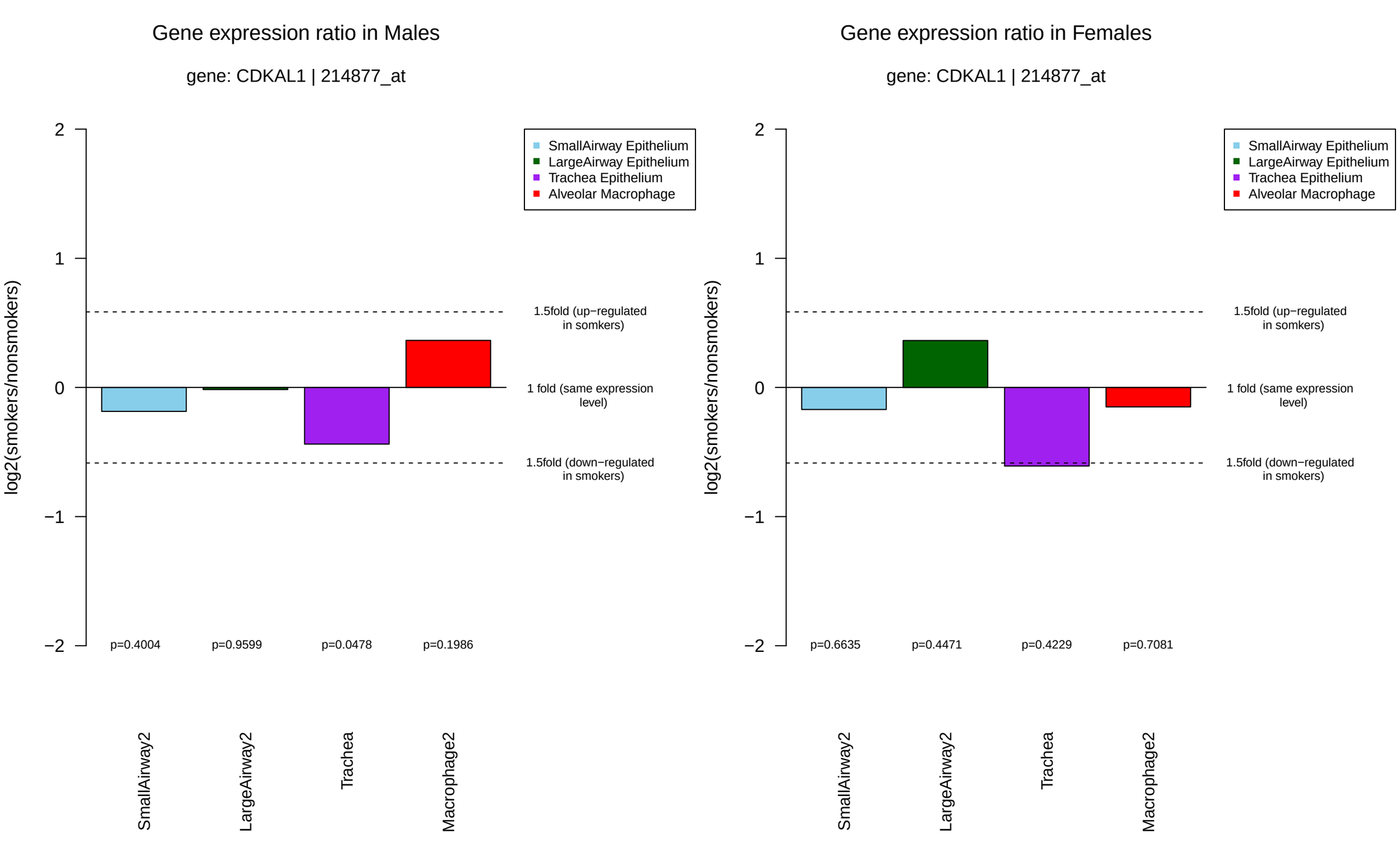
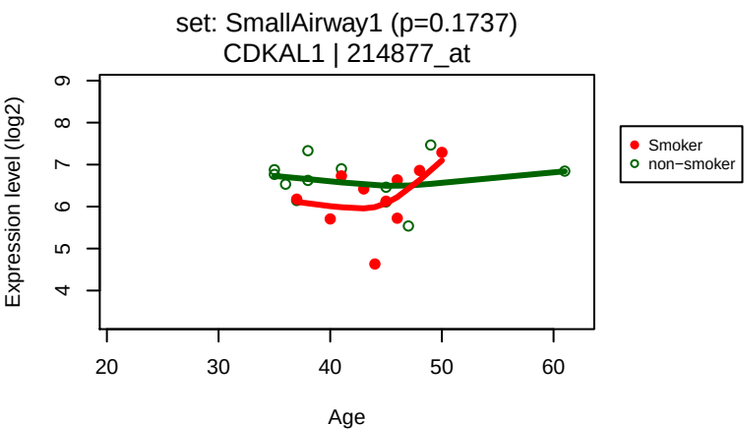
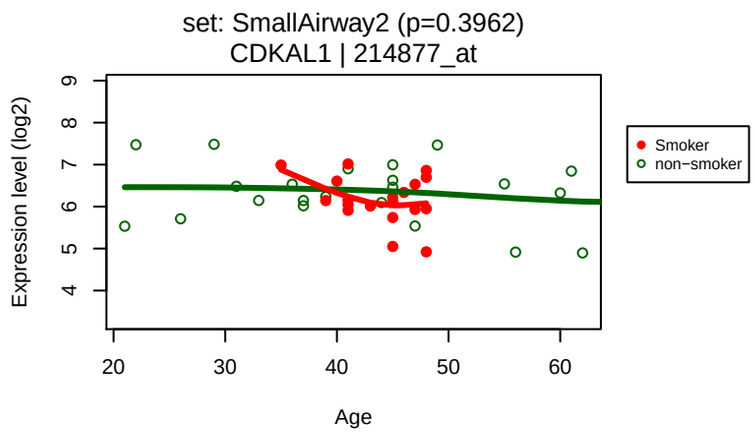
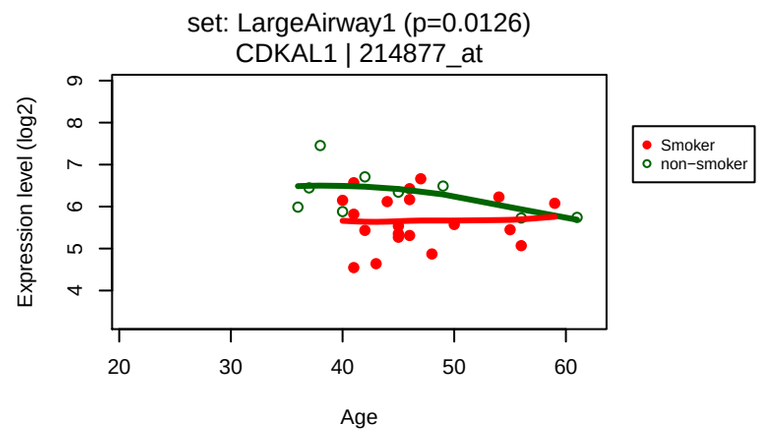

Gene expression of smokers and nonsmokers
gene: CDKAL1 | 214877_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1737
p=0.3962
p=0.1264
p=0.0126
p=0.6941
p=0.0163
p=0.3162
p=0.3799
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

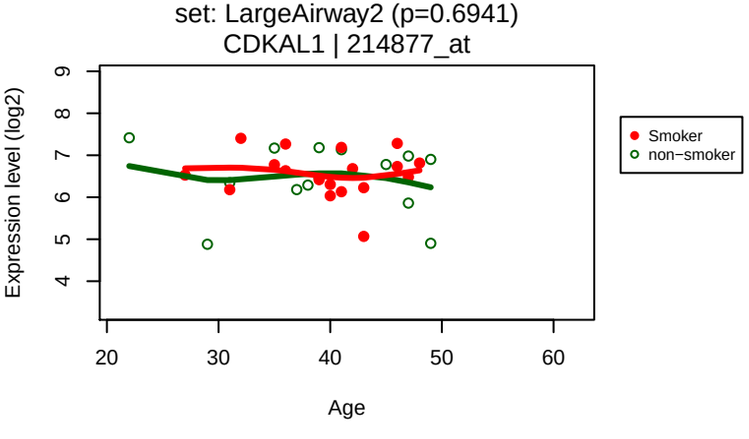

Gene expression ratio (smokers/nonsmokers)
gene: CDKAL1 | 214877_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1737
p=0.3962
p=0.1264
p=0.0126
p=0.6941
p=0.0163
p=0.3162
p=0.3799
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

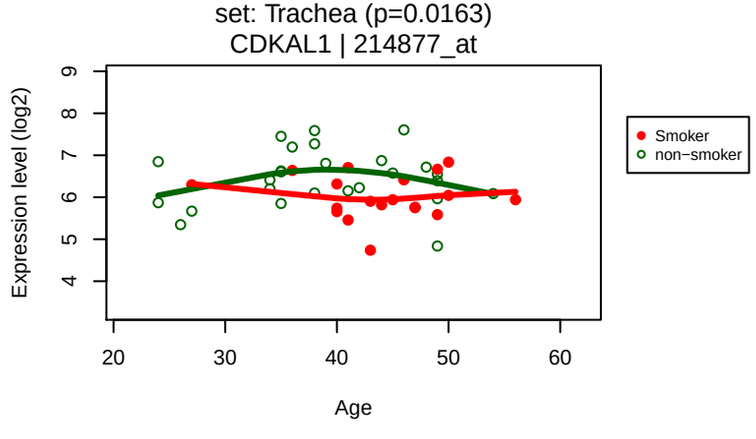

Gene expression ratio in Males
gene: CDKAL1 | 214877_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4004
p=0.9599
p=0.0478
p=0.1986
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CDKAL1 | 214877_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6635
p=0.4471
p=0.4229
p=0.7081
SmallAirway2
LargeAirway2
Trachea
Macrophage2

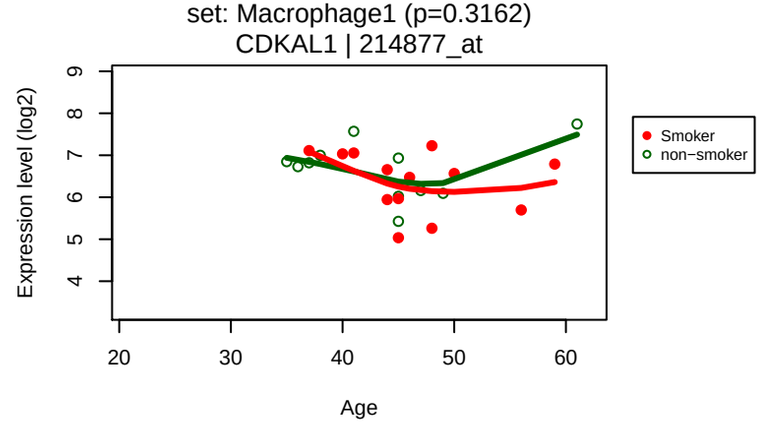

set: SmallAirway1 (p=0.1737)
CDKAL1 | 214877_at
Expression level (log2)
Age
Smoker
non−smoker

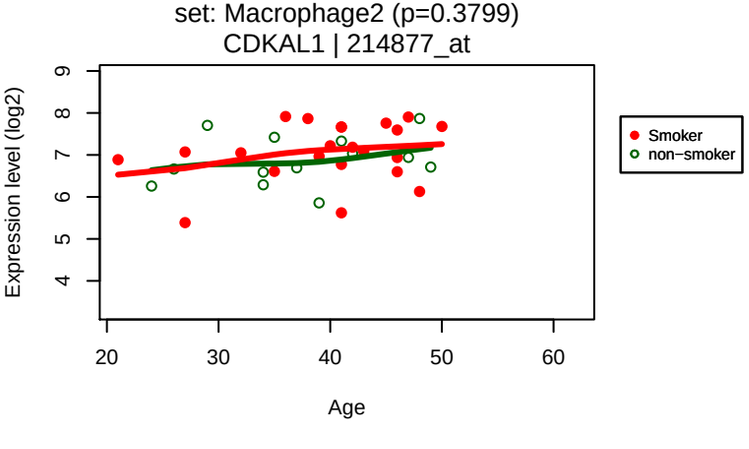

set: SmallAirway2 (p=0.3962)
CDKAL1 | 214877_at
Expression level (log2)
Age
Smoker
non−smoker

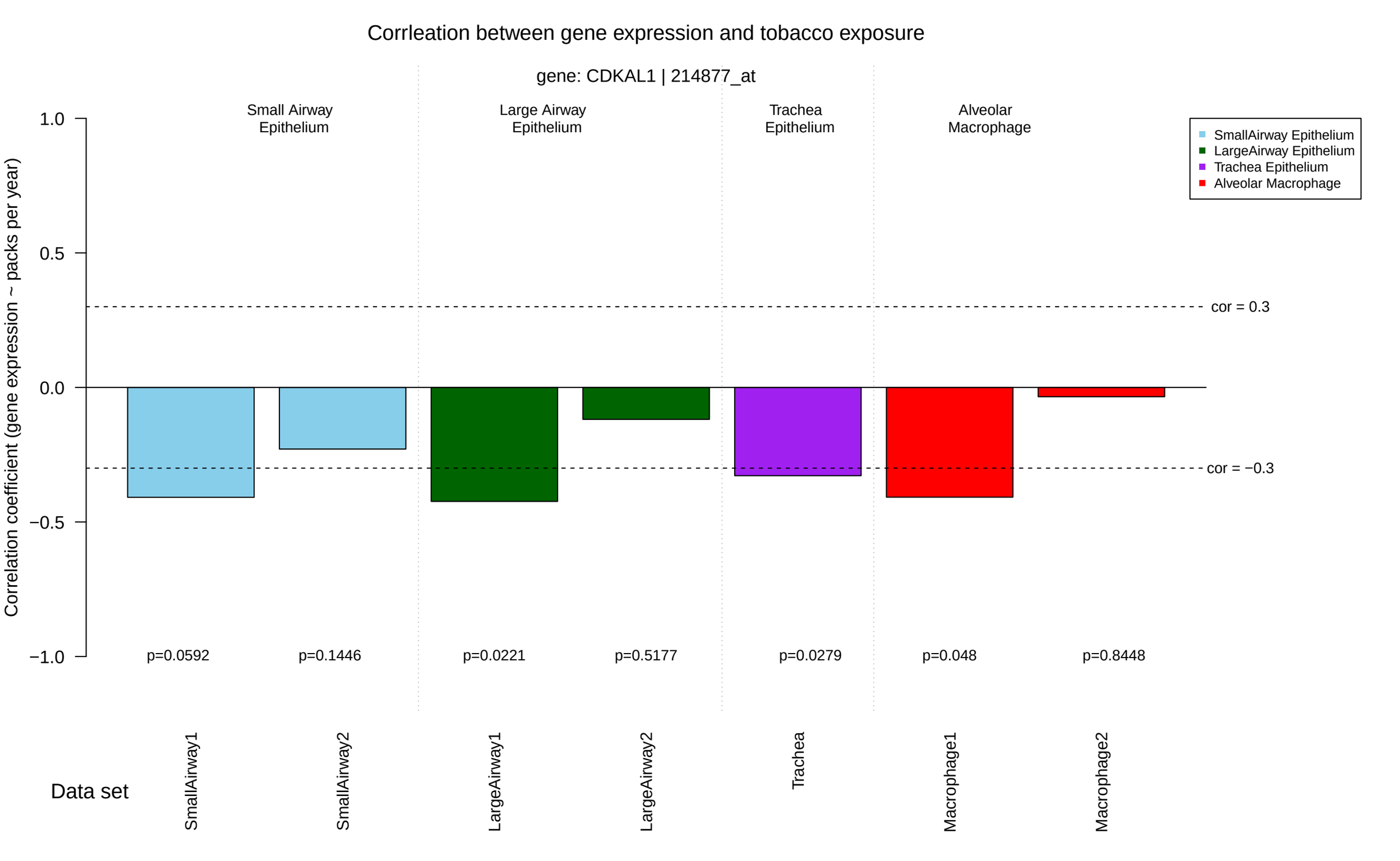

set: LargeAirway1 (p=0.0126)
CDKAL1 | 214877_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6941)
CDKAL1 | 214877_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0163)
CDKAL1 | 214877_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3162)
CDKAL1 | 214877_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3799)
CDKAL1 | 214877_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CDKAL1 | 214877_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0592
p=0.1446
p=0.0221
p=0.5177
p=0.0279
p=0.048
p=0.8448
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2